Zomato Restaurant Data Analysis

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Tools Used: Python, SQL, Excel, Power BI
Date: 2025

Executive Summary:

This report presents a comprehensive analysis of restaurant data obtained from the Zomato platform with the objective of understanding restaurant distribution, customer preferences, pricing trends, and service performance. The analysis was conducted using Python, SQL, Excel, and Power BI to ensure accurate data cleaning, in-depth exploration, and interactive visualization.

The dataset includes key attributes such as restaurant location, cuisine type, customer ratings, votes, average cost for two, online ordering availability, and table booking services. A structured data-cleaning process was applied to remove duplicates, handle missing values, and standardize categorical fields to improve data quality and reliability.

Exploratory data analysis reveals that restaurant concentration is highest in central and high-traffic locations, indicating strong competition in these areas. Restaurants offering online ordering and table booking services consistently show higher customer ratings and engagement, highlighting the importance of service convenience. Additionally, mid-range priced restaurants demonstrate better overall performance compared to low-cost and premium segments, suggesting an optimal balance between affordability and quality.

Interactive dashboards developed in Excel and Power BI allow stakeholders to filter and analyse restaurant performance by location, service type, and cost range, enabling data-driven decision-making. These dashboards provide quick insights into key performance indicators such as average ratings, total votes, and restaurant count.

Based on the findings, the report recommends focusing on expanding online ordering services, optimizing pricing strategies within the mid-range segment, and targeting high-performing locations for business expansion. Overall, this analysis delivers actionable insights that can support strategic planning, improve customer satisfaction, and enhance operational efficiency in the restaurant business.

Business Objectives:

The primary objective of this analysis is to derive actionable business insights from Zomato restaurant data that can support strategic decision-making for restaurant owners, investors, and food service platforms. The specific business objectives of this study are outlined below:

1. Analyse Restaurant Distribution by Location. To identify areas with high and low restaurant density in order to understand market competition and potential opportunities for expansion.
2. Evaluate Customer Ratings and Engagement. To assess rating patterns and vote counts across restaurants and determine the factors that influence customer satisfaction and engagement.
3. Assess the Impact of Online Ordering Services. To examine whether restaurants offering online ordering receive higher ratings and greater customer interaction compared to those that do not.

4. Analyse the Effect of Table Booking Availability. To understand how table booking services impact customer preferences, ratings, and overall restaurant performance.
5. Examine Pricing Trends and Performance. To evaluate the relationship between average cost for two and restaurant ratings in order to identify optimal pricing ranges.
6. Identify Popular Cuisines and Service Combinations. To determine the most preferred cuisines and analyse how different service combinations (online order and table booking) influence performance.
7. Support Data-Driven Business Decisions. To provide insights that can help stakeholders optimize service offerings, improve customer experience, and plan future growth strategies.

Data Overview:

The analysis is based on a structured dataset obtained from the Zomato platform, which contains detailed information about restaurants across multiple locations. The dataset is designed to capture key aspects of restaurant performance, customer engagement, pricing, and service availability.

The dataset includes records for several thousand restaurants and consists of multiple attributes such as restaurant name, location, cuisine type, customer ratings, total votes, average cost for two people, online ordering availability, and table booking options. These variables collectively provide a comprehensive view of how restaurants operate and perform in different market segments.

Each record in the dataset represents a unique restaurant and includes both numerical and categorical fields. Numerical fields such as ratings, votes, and average cost enable quantitative analysis, while categorical fields such as location, cuisine, online ordering, and table booking allow for segmentation and comparative analysis.

The data serves as the foundation for exploratory data analysis, dashboard creation, and insight generation. Before analysis, the dataset was carefully reviewed and cleaned to ensure consistency, accuracy, and reliability. This ensured that all subsequent findings and business recommendations were based on high-quality and trustworthy data.

Data Cleaning Process:

Before performing analysis and building dashboards, the raw Zomato dataset was thoroughly cleaned and prepared to ensure accuracy, consistency, and reliability of the results. Since real-world data often contains inconsistencies and missing values, a structured data-cleaning approach was followed using Python, SQL, and Excel.

1. Handling Missing and Null Values:

The dataset was examined for missing and null values across all columns. Records with critical missing information such as ratings or location were removed, while non-critical missing values were handled appropriately to avoid distortion in the analysis.

2. Removing Duplicate Records:

Duplicate restaurant entries were identified based on restaurant name and location. These duplicate records were removed to ensure that each restaurant was represented only once in the dataset, preventing overcounting and biased results.

3. Standardizing Data Formats:

Several columns contained inconsistent formatting. Customer ratings were converted into a uniform numerical format, and categorical variables such as online ordering and table booking were standardized to consistent values (e.g., Yes/No). Text fields were cleaned to remove extra spaces and inconsistencies.

4. Cleaning and Validating Categorical Data:

Invalid or irrelevant values in categorical columns were filtered out. This ensured that categories such as location, cuisine, and service availability were accurate and suitable for grouping and comparison during analysis.

5. Outlier Detection and Filtering:

Extreme or unrealistic values in numerical fields such as average cost were reviewed and filtered where necessary to maintain realistic and meaningful analysis results.

6. Column Selection and Optimization:

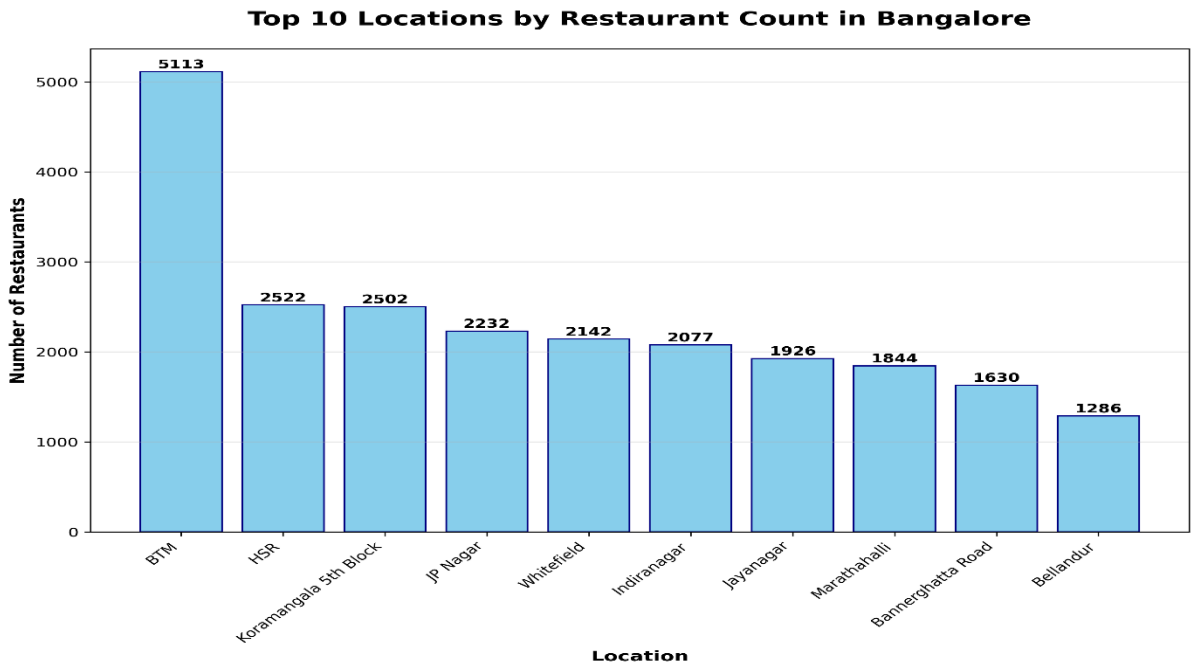
Irrelevant or unused columns were removed to streamline the dataset and improve analysis efficiency. Only business-relevant fields were retained for exploratory analysis and dashboard development.

7. Final Data Validation:

After cleaning, the dataset was revalidated to confirm consistency across all fields. Summary statistics and sample checks were performed to ensure that the cleaned data was ready for accurate analysis and visualization.

Exploratory Data Analysis (EDA):

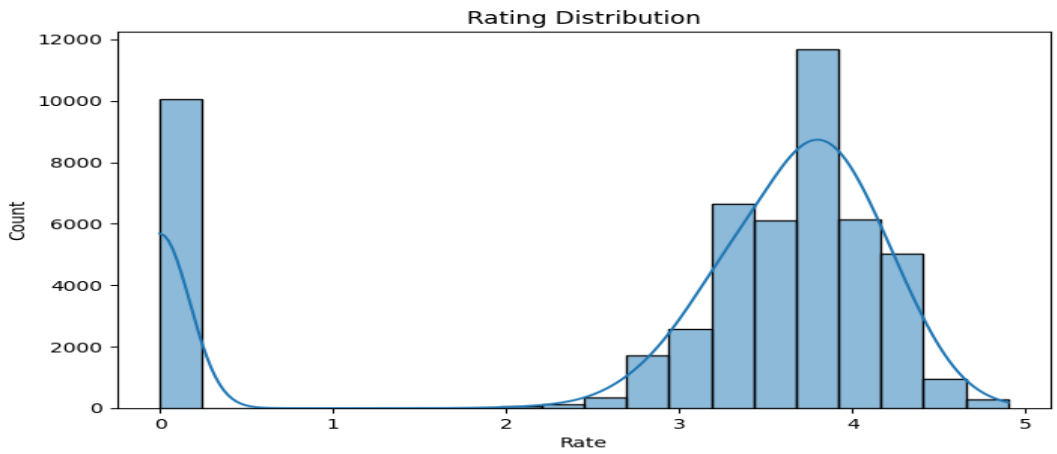
1. Restaurant Count by Location:



Explanation:

This chart shows the Top 10 restaurants across different locations. BTN, HSR and Koramangala 5th Block locations have a higher concentration of restaurants, indicating strong demand but increased competition. Locations with fewer restaurants may present potential opportunities for expansion.

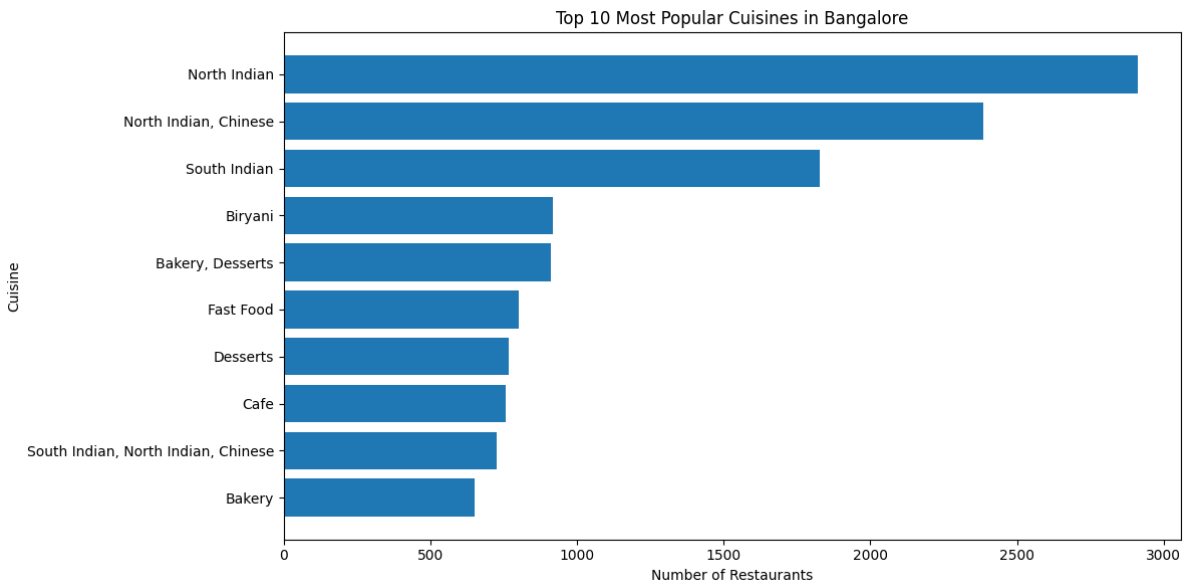
2. Rating Distribution



Explanation:

This chart shows Most restaurants are rated between 3.0 and 4.5, with the highest concentration around 3.5–4.0, indicating that customers generally provide positive reviews. There is a noticeable spike at a 0 rating, which likely represents missing, unrecorded, or invalid ratings rather than actual customer feedback. The distribution is right-skewed due to the spike at zero but generally resembles a normal distribution for valid ratings (3–5). This suggests most restaurants perform consistently well.

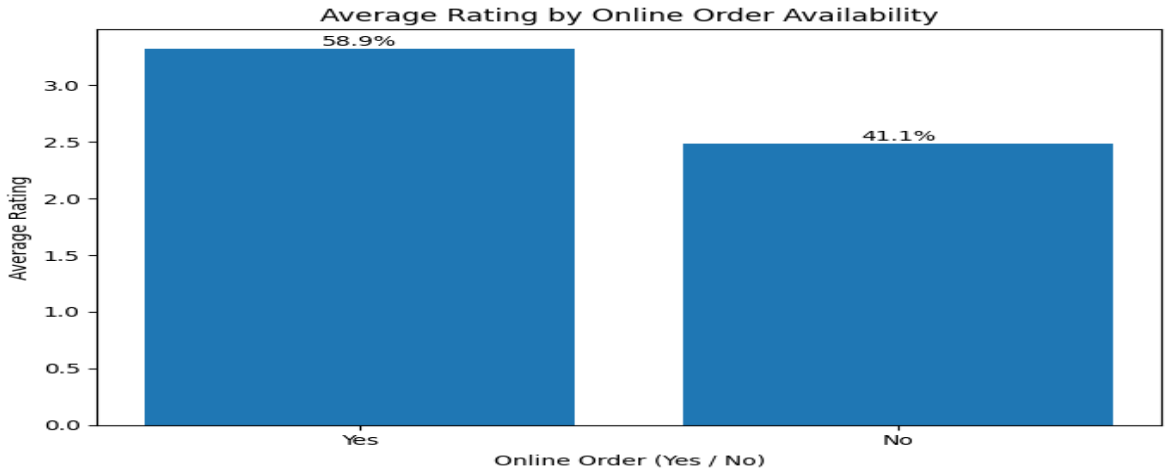
3. Top Cuisines by Restaurant Count:



Explanation:

A small number of cuisines dominate the market in terms of restaurant count. Popular cuisines attract higher customer engagement, making them key drivers of restaurant visibility and demand.

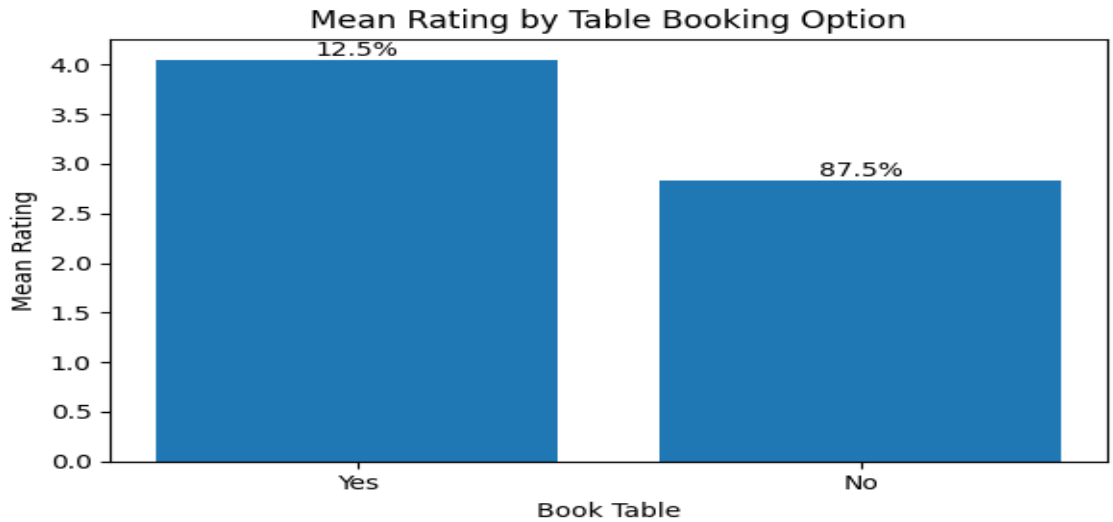
4. Online Ordering vs Average Rating:



Explanation:

Restaurants offering online ordering show higher average ratings compared to those without this service. This highlights the importance of convenience in improving customer satisfaction.

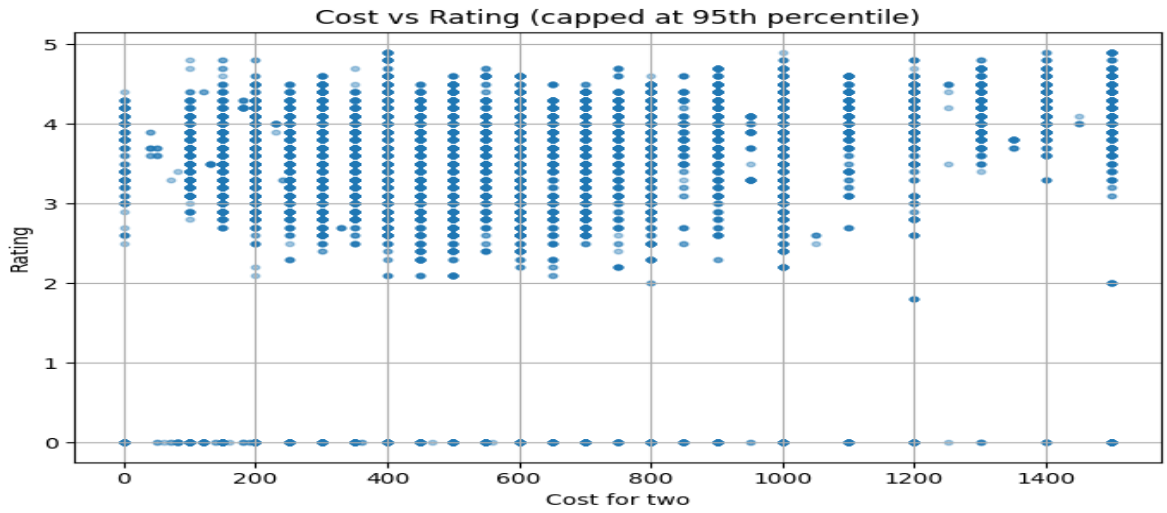
5. Table Booking vs Average Rating:



Explanation:

Restaurants with table booking facilities tend to receive better ratings than those without. This trend is more noticeable in premium and high-traffic locations where advance reservations enhance the dining experience.

6. Cost vs Rating:



Explanation:

Mid-range priced restaurants achieve higher ratings compared to low-cost and high-cost segments. This indicates that customers prefer a balance between price and quality.

8. Service Combination Impact (Online Order + Table Booking):

Explanation:

Restaurants offering both online ordering and table booking services achieve the highest average ratings. Combined services significantly improve customer experience and engagement.

Dashboard Explanation:

To present the analysis in an interactive and user-friendly manner, dashboards were developed using Microsoft Excel and Power BI. These dashboards allow stakeholders to explore restaurant performance dynamically and gain quick insights without needing technical knowledge.

1. Excel Dashboard Overview:

The Excel dashboard was designed to provide a high-level overview of restaurant performance using PivotTables, Pivot Charts, and slicers. It includes key visualizations such as restaurant count by location, average ratings by service type, cost distribution, and cuisine analysis. Slicers enable users to filter data by location, online ordering availability, and table booking options, allowing for flexible and customized analysis.

This dashboard is particularly useful for quick analysis and operational decision-making. Users can easily compare performance across different segments and identify trends using simple interactive controls.

2. Power BI Dashboard Overview:

The Power BI dashboard offers a more advanced and visually rich analysis experience. It includes key performance indicator (KPI) cards displaying total restaurants, average rating, total votes, and average cost. Interactive charts and graphs provide insights into location-wise performance, service impact, pricing trends, and customer engagement.

Advanced filtering and drill-down features allow users to explore data at multiple levels, such as moving from city-level insights to individual locations or service types. This enables deeper analysis and supports strategic decision-making.

3. Interactivity and User Controls:

Both dashboards are equipped with filters and slicers that allow users to:

- Analyse restaurant performance by location
- Compare online ordering versus non-online ordering restaurants
- Evaluate the impact of table booking services
- Examine pricing segments and rating trends

These interactive features empower users to customize views based on their business needs and extract insights relevant to specific scenarios.

4. Business Value of the Dashboards:

The dashboards transform complex datasets into clear visual stories. They enable stakeholders to quickly identify high-performing locations, understand customer preferences, and evaluate service effectiveness. By providing real-time, interactive insights, the dashboards support data-driven decisions that can improve customer experience, optimize pricing strategies, and guide business expansion plans.

Key Insights:

Based on the exploratory data analysis and dashboard insights, several important patterns and trends were identified that provide valuable understanding of restaurant performance and customer behaviour.

1. **High Restaurant Concentration in Central Locations:**

   The analysis shows that central and high-traffic locations have the highest number of restaurants. While these areas offer strong customer demand, they also experience intense competition. Restaurants operating in such locations need differentiated offerings to stand out.

2. **Positive Impact of Online Ordering Services**:

   Restaurants that provide online ordering consistently demonstrate higher average ratings and increased customer engagement compared to those that do not. This indicates that convenience and accessibility play a significant role in improving customer satisfaction.

3. **Table Booking Enhances Customer Engagement**:

   Restaurants offering table booking services tend to receive more votes and better ratings, especially in premium and busy locations. This suggests that advance reservation options improve the overall dining experience and customer trust.

4. **Mid-Range Pricing Performs Best**:

   The relationship between average cost and ratings reveals that mid-range priced restaurants generally receive higher ratings than both low-cost and high-cost establishments. Customers appear to value a balance between price and quality.

5. **Cuisine Preferences Influence Popularity**:

   A limited number of cuisines dominate the market and attract higher customer engagement. Restaurants offering popular cuisines benefit from greater visibility and stronger customer interest.

6. **Customer Engagement Correlates with Service Quality**:

   Higher vote counts are strongly associated with better ratings and the availability of services such as online ordering and table booking. This highlights the importance of service quality in driving customer interaction.

7. **Opportunities in Low-Density Locations**:

   Locations with fewer restaurants but reasonable customer engagement present potential opportunities for business expansion with comparatively lower competition.

Business Recommendations:

Based on the key insights and findings from the analysis, the following business recommendations are proposed to help improve restaurant performance, enhance customer satisfaction, and support data-driven decision-making.

1. **Expand and Strengthen Online Ordering Services**:

   Restaurants should prioritize offering and optimizing online ordering services, as data shows a strong positive impact on customer ratings and engagement. Improving delivery speed, packaging quality, and menu visibility on digital platforms can further enhance customer experience.

2. **Promote Table Booking in High-Traffic Locations:**

   Table booking services should be actively promoted, particularly in premium and high-demand areas. Advance reservations improve customer convenience, reduce waiting time, and contribute to higher ratings and repeat visits.

3. **Focus on Optimal Pricing Strategies:**

   Restaurants should consider positioning their pricing within the mid-range segment, which demonstrates the strongest performance in terms of customer ratings. Aligning price with perceived quality can increase customer satisfaction and loyalty.

4. **Target Expansion in Low-Competition Locations**:

   Locations with lower restaurant density but steady customer engagement represent potential opportunities for expansion. Entering these markets can help reduce competitive pressure while capturing unmet demand.

5. **Leverage Popular Cuisines for Market Growth:**

   Businesses should analyse cuisine popularity and focus on high-demand cuisine categories when planning new outlets or updating menus. This can increase visibility and attract a larger customer base.

6. **Improve Customer Experience to Drive Engagement:**

   Enhancing service quality, cleanliness, and overall dining experience can significantly increase customer ratings and vote counts. Positive customer experiences directly translate into better online visibility and brand reputation.

7. **Use Dashboards for Continuous Monitoring**:

   Stakeholders should regularly use the Excel and Power BI dashboards to monitor performance metrics such as ratings, votes, and service adoption. Continuous data monitoring enables timely decisions and quick response to changing market trends.

Conclusion:

While the analysis provides valuable insights into restaurant performance and customer behaviour, it is important to consider certain limitations associated with the dataset and the analytical approach.

1. **Data Scope and Coverage**:

   The dataset represents a limited snapshot of restaurants available on the Zomato platform and may not include all restaurants across every location. As a result, findings may not fully reflect the entire restaurant market.

2. **Time Dependency of Data**:

   The data does not capture changes over time such as seasonal trends, recent restaurant openings, or updated customer ratings. Business dynamics may have evolved since the data was collected.

3. **Subjectivity of Customer Ratings**:

   Restaurant ratings and reviews are subjective and influenced by individual customer preferences. Factors such as personal taste, expectations, and isolated experiences can affect ratings and may not always represent overall service quality.

4. **Limited Operational Details**:

   The dataset does not include operational factors such as service speed, staff quality, marketing efforts, or promotional activities, which can also impact customer satisfaction and performance.

5. **Potential Data Quality Issues**:

   Despite thorough data cleaning, minor inaccuracies or inconsistencies may still exist due to manual entries or platform-level limitations.

6. **Platform-Specific Insights**:

   The analysis is based solely on Zomato data. Customer behaviour and performance trends may vary across other food delivery or restaurant discovery platforms.

This analysis of Zomato restaurant data provides a comprehensive understanding of restaurant performance, customer preferences, and service trends across different locations. By applying systematic data cleaning, exploratory data analysis, and interactive dashboarding techniques, meaningful insights were derived to support informed business decision-making.

The findings highlight the importance of service convenience, particularly online ordering and table booking, in driving higher customer ratings and engagement. Additionally, pricing analysis indicates that mid-range restaurants tend to achieve better customer satisfaction, reflecting a strong relationship between perceived value and quality. Location-based insights further reveal areas of high competition as well as potential opportunities for expansion in lower-density markets.

The Excel and Power BI dashboards developed as part of this project enable stakeholders to explore these insights interactively and monitor key performance indicators with ease. These tools transform complex data into actionable intelligence that can guide strategic planning and operational improvements.

Overall, this report demonstrates how data-driven analysis can enhance customer experience, optimize pricing strategies, and support sustainable business growth. The insights and recommendations presented can serve as a foundation for future analysis and continuous performance monitoring as market conditions evolve.